

CHRISTIAN B. HUGHES

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EDUCATION

Czech Technical University in Prague

B.S. in Theoretical Informatics

Prague, Czech Republic

Aug 2023 - Present

- **Related Coursework:** Graph Theory, Real & Complex Analysis, Measure Theory, Statistics & Applications, Complexity Analysis of Algorithms, Functional Programming

University of Central Florida

Studies in Mechanical Engineering

Orlando, Florida

Aug 2022 - May 2023

PUBLICATIONS

Francesco Dolce, Christian B. Hughes. *Clustering of Return Words in Languages of Interval Exchanges. Lecture Notes in Computer Science*, vol. 15729, Springer. 2025

Francesco Dolce, Christian B. Hughes. *Extended Branching Rauzy Induction*. Preprint. 2025

RESEARCH TALKS

Bratteli-Vershik Models from Branching Rauzy Induction. Dyadisc9, Saint-Valery-sur-Somme, France. 2026

Induced Dynamics of Interval Exchange Transformations. TIGR CoW 2026, Janov nad Nisou, Czechia. 2026

EXPERIENCE

IEAP Prague + CERN ATLAS

Machine Learning Research Assistant

Prague, Czech Republic

Feb 2025 - Present

- Developing novel approaches to Higgs boson mass reconstruction using ML models
- Collaboratively applying cutting-edge techniques in machine learning, including genetic algorithms and physics-informed neural networks

Northrop Grumman

Engineering Intern

St. Augustine, Florida

Oct 2020 - May 2022

- Engaged in the group development of pneumatic exoskeleton legs for use in industrial settings
- Programmed an ARM-based microcontroller in C to control solenoid systems for pneumatic actuation
- Presented project progress to an audience of facility engineers and managed the budget for exoskeleton development

TEACHING

Teaching Assistant, Czech Technical University in Prague

Core mathematics courses (Linear Algebra I/II, Mathematical Analysis I/II)

Prague, Czech Republic

Fall 2024 - Present

- Led weekly problem and theory sessions, including board proofs, worked examples, and exam reviews.
- Courses: Linear Algebra I (Fall 2025, Fall 2024), Linear Algebra II (Spring 2025, Spring 2026), Mathematical Analysis I (Spring 2025, Spring 2026), Mathematical Analysis II (Fall 2025).

SKILLS

Research Interests: Symbolic dynamics, Combinatorics on words, Interval exchange transformations, operator algebras

Programming: Python, C++, C, MATLAB

Tools: L^AT_EX, Git, Numpy

Languages: English (Native), Czech (B2), Spanish (A2)